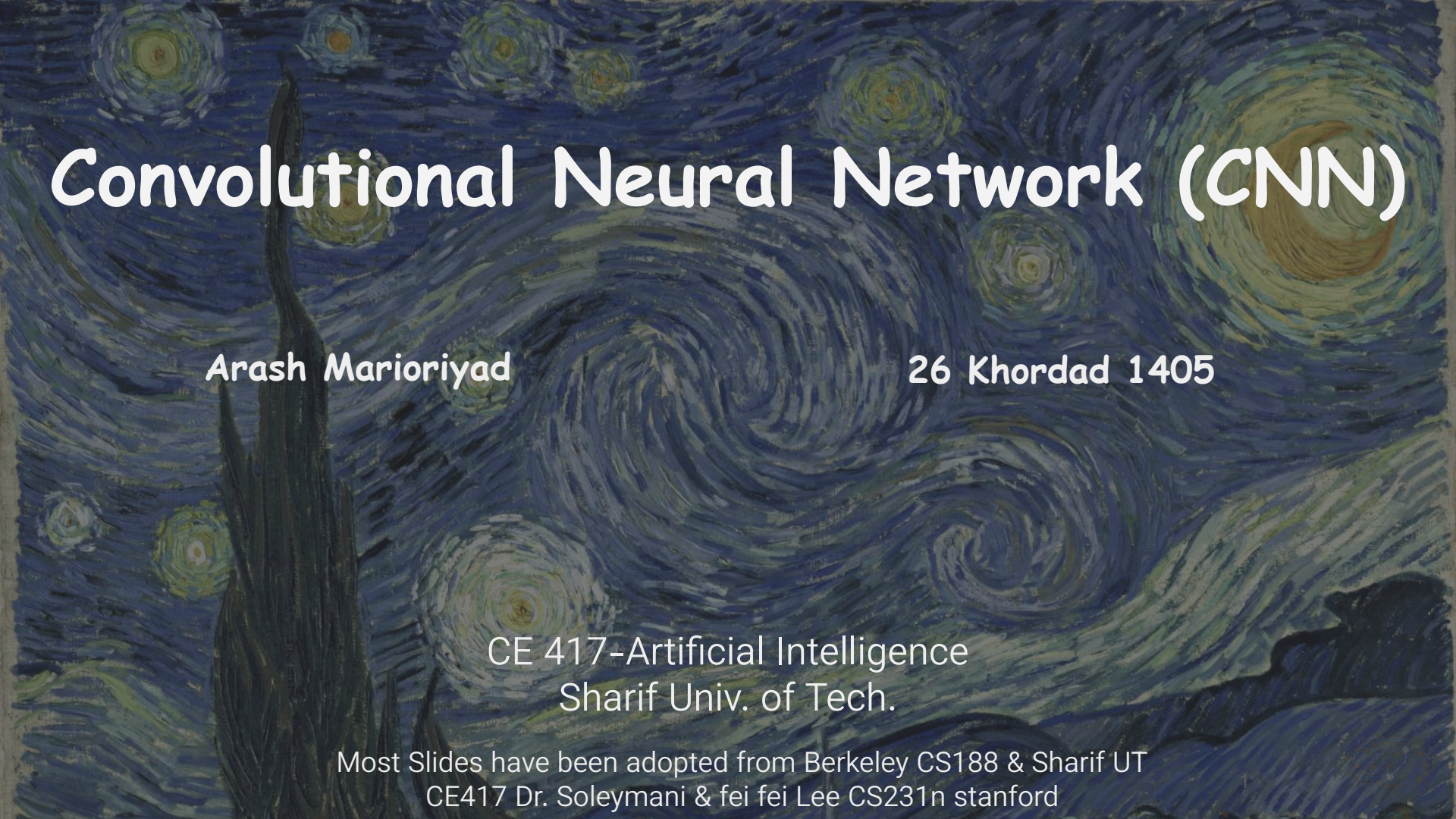


The background of the slide is a reproduction of the painting 'The Starry Night' by the Dutch Impressionist painter J.M.W. Turner. The painting depicts a turbulent, swirling night sky with a bright, glowing yellow sun or moon in the upper right corner. The sky is filled with numerous smaller, swirling stars and nebulae. In the foreground, a dark, silhouetted cypress tree stands on the left, and a small, dark, rocky island is visible in the lower right. The overall color palette is dominated by deep blues, greens, and yellows.

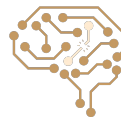
Convolutional Neural Network (CNN)

Arash Marioriyad

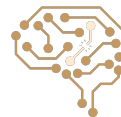
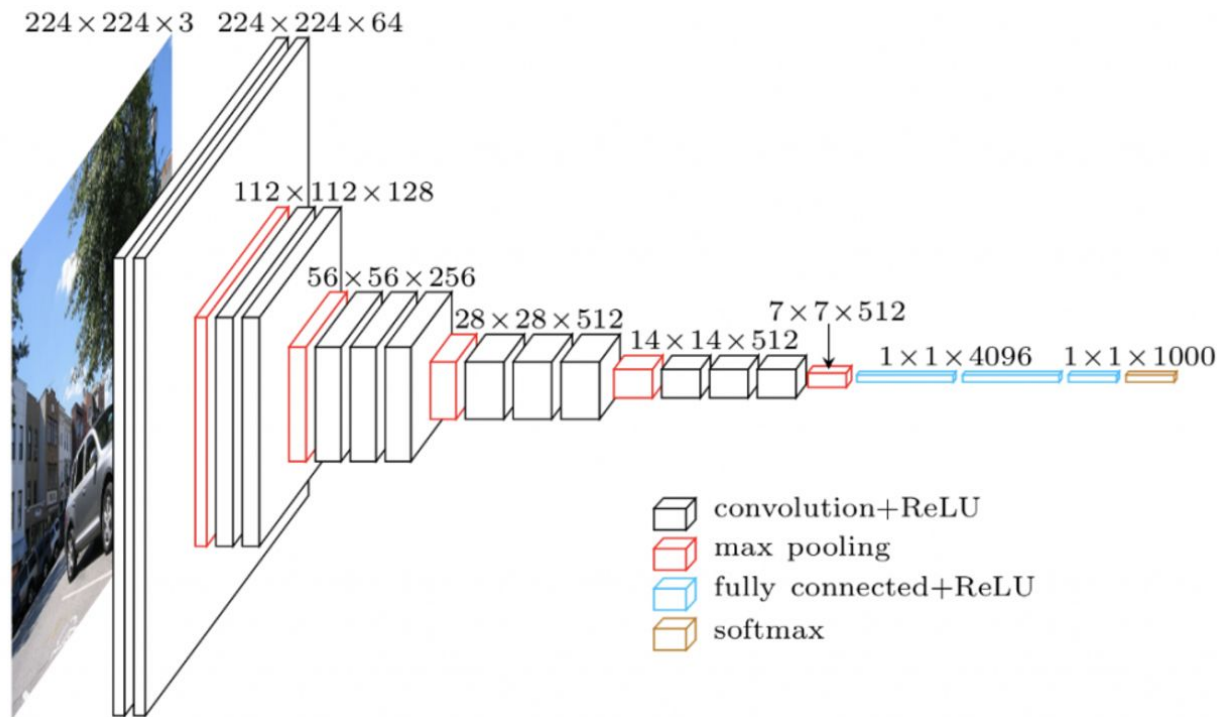
26 Khordad 1405

CE 417-Artificial Intelligence
Sharif Univ. of Tech.

Most Slides have been adopted from Berkeley CS188 & Sharif UT
CE417 Dr. Soleymani & fei fei Lee CS231n stanford

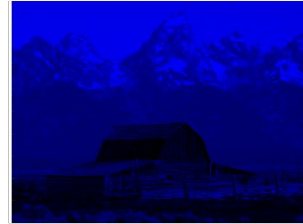
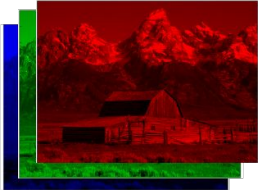


Convolutional Neural Networks



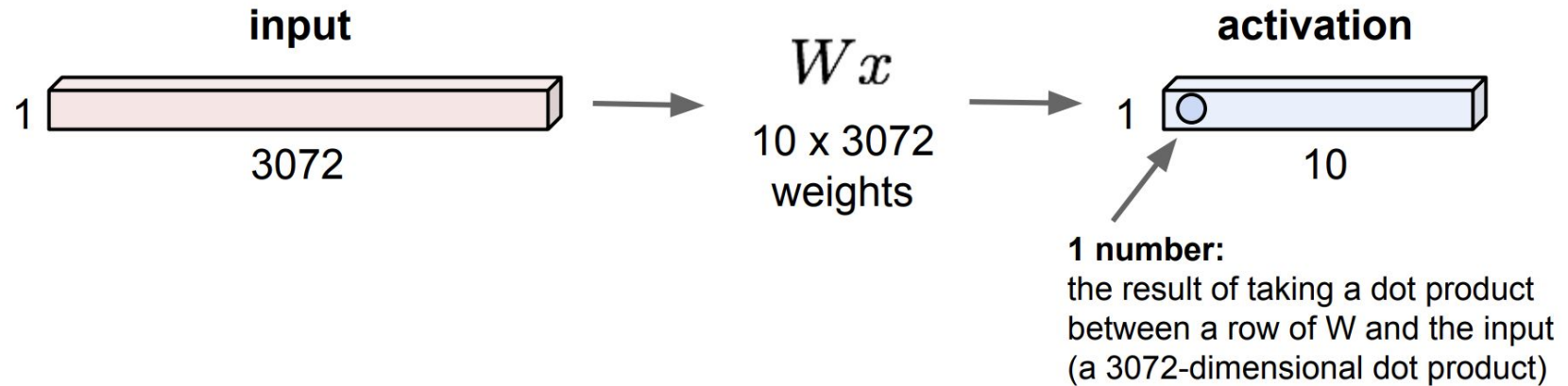
Colored Images: RGB

- Image Shape: $H \times W \times 3$



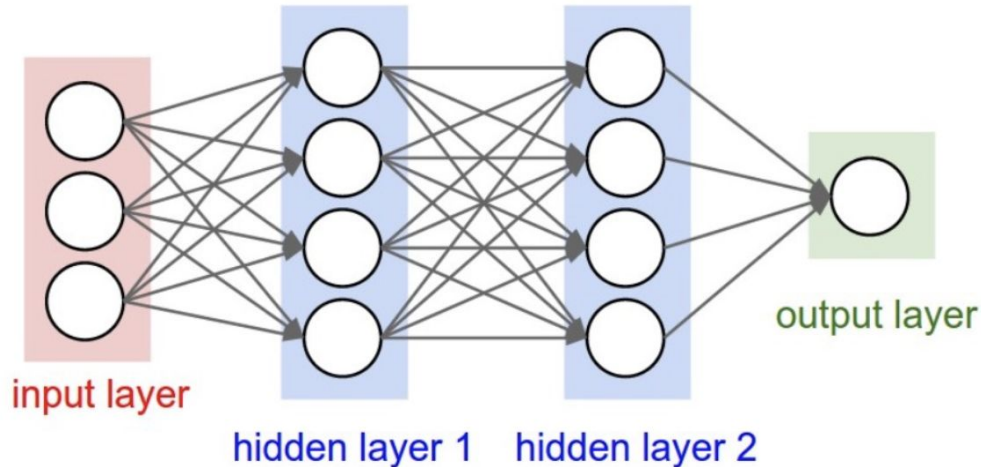
First Idea: Fully Connected Layer

32x32x3 image -> stretch to 3072 x 1



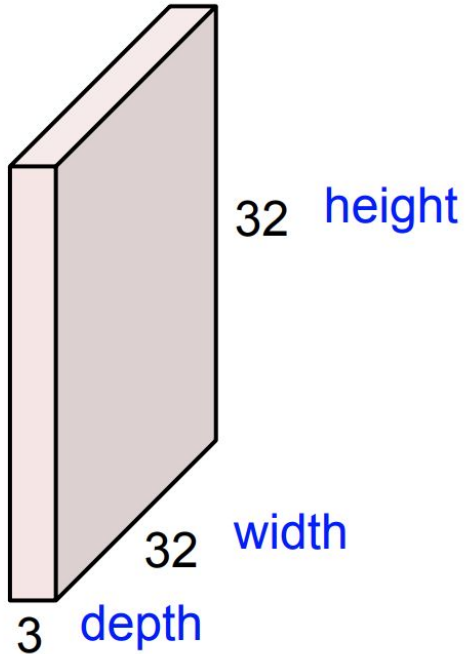
Problem of Fully Connected Network for Image as Input

- Requires $32 \times 32 \times 3 = 3072$ weights just for a single neuron in the 2nd layer.
- For larger images, number of weights increases rapidly, leading to overfitting.

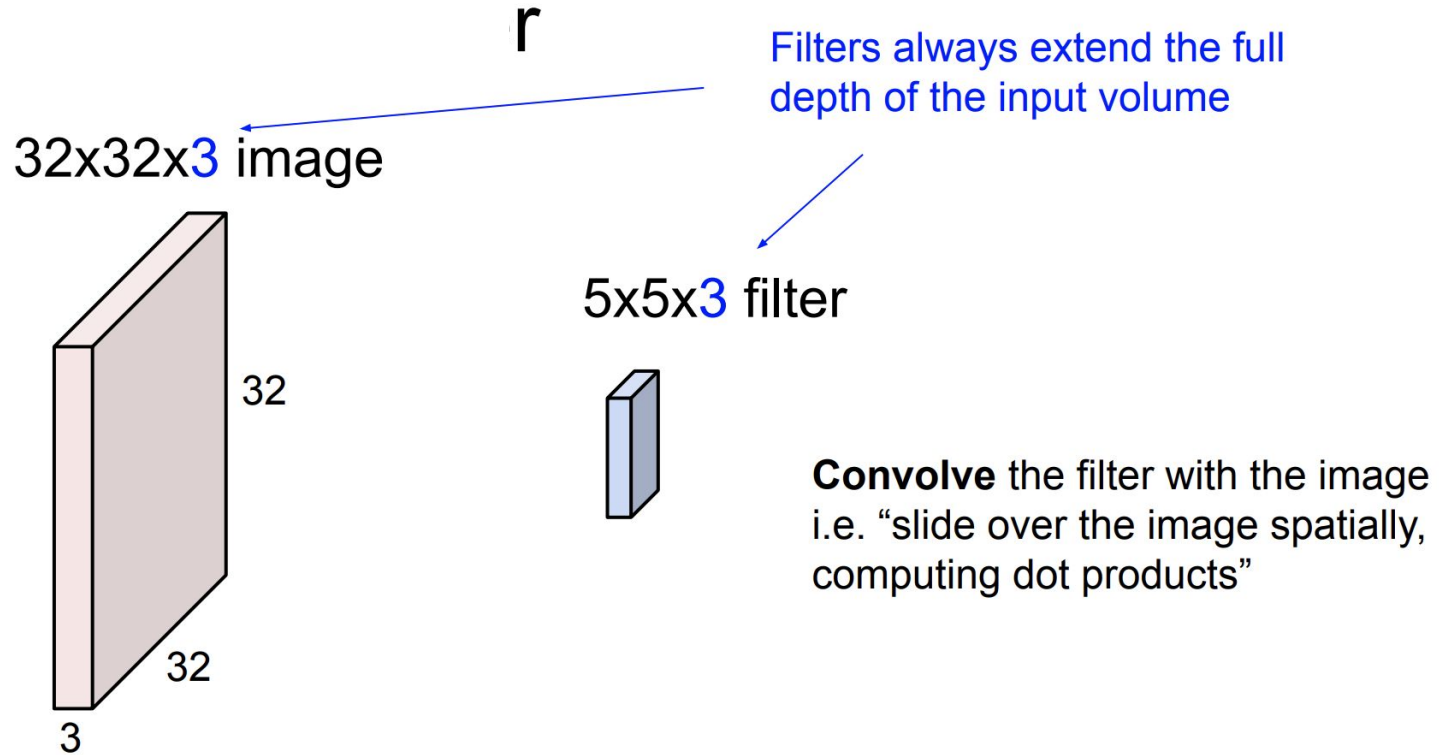


Convolution Layer

32x32x3 image -> preserve spatial structure



Convolution Layer: Filter=Weights

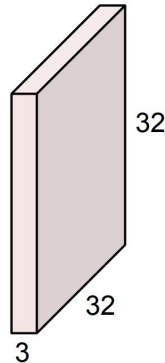


Convolution Layer

- Each neuron in the hidden layers is **only connected to a few number of neurons** in the previous layer.

Conv

32x32x3 image



Filters always extend the full depth of the input volume

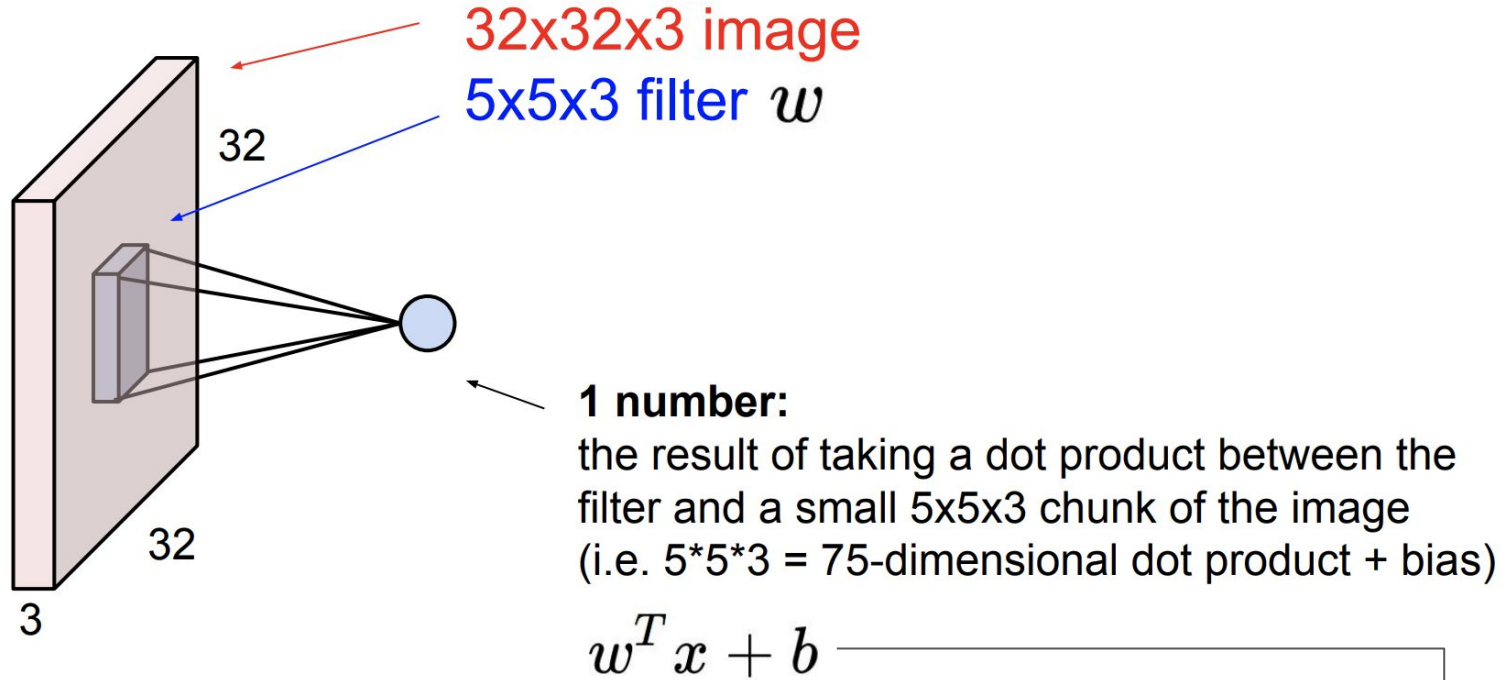
5x5x3 filter



Convolve the filter with the image
i.e. “slide over the image spatially,
computing dot products”



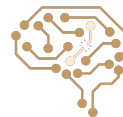
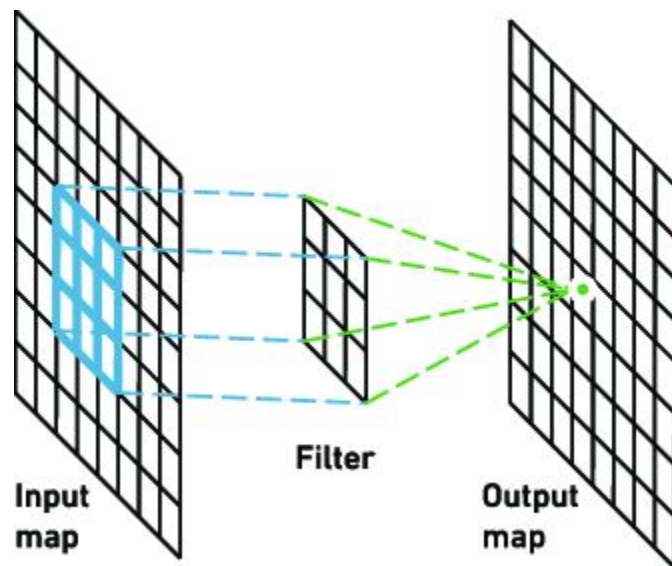
Convolution Layer



Then, apply activation function on the above dot product + bias

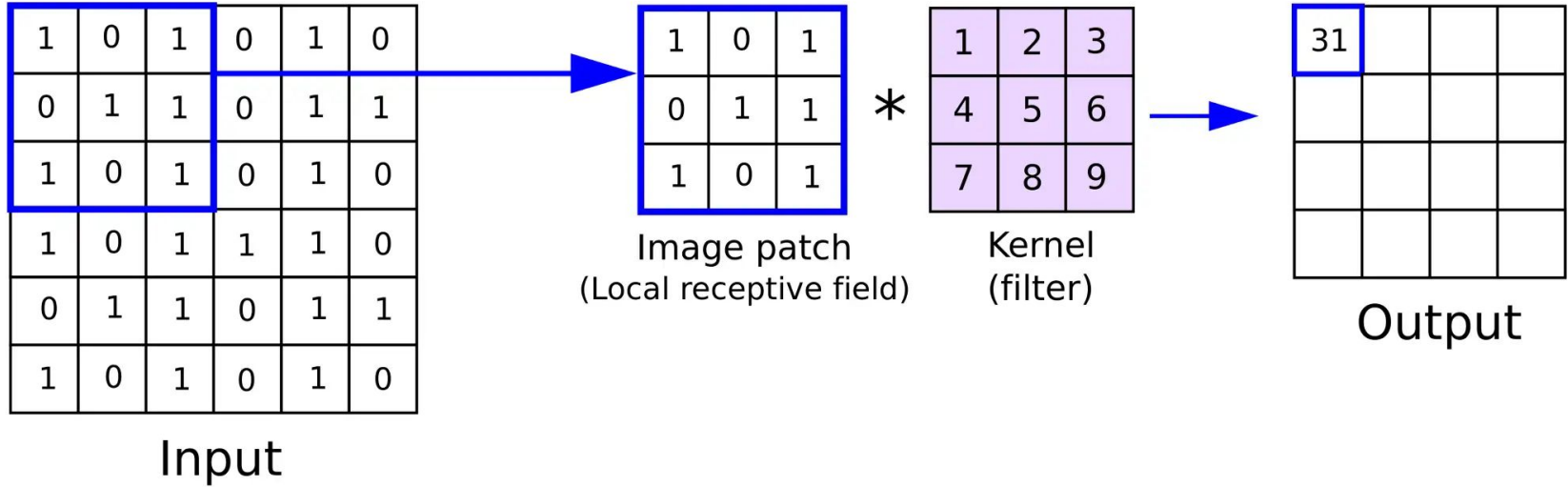


Convolution Layer

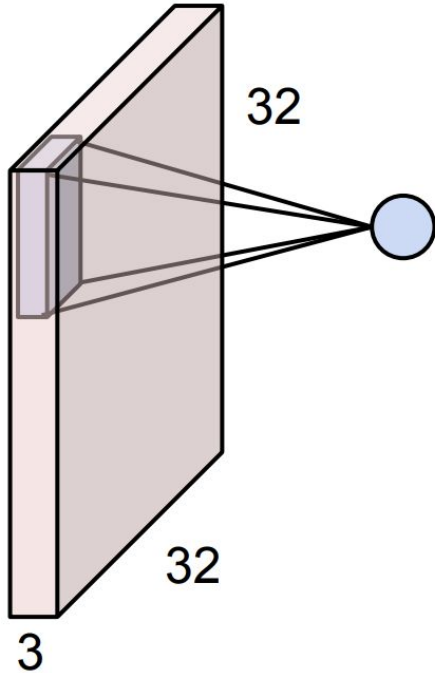


Convolution Layer

iq.opengenus.org



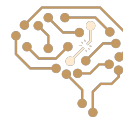
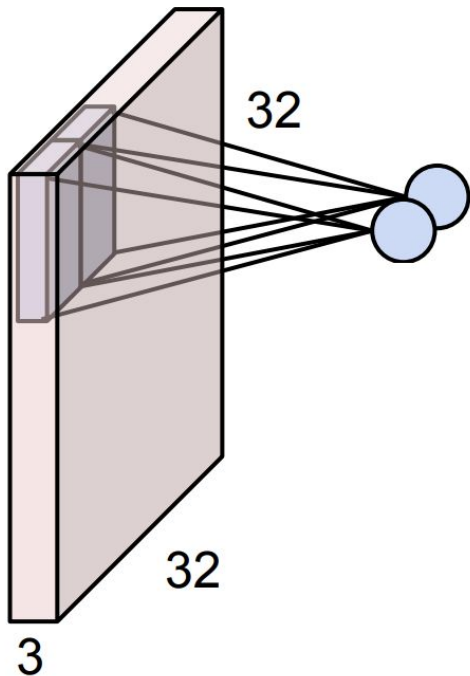
Convolution Layer



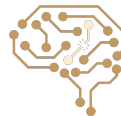
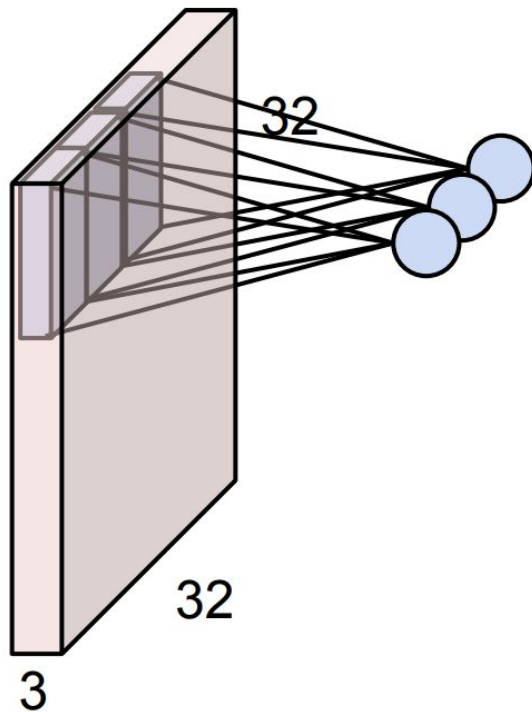
Align with human's inductive bias, as each filter searches for a specific pattern in all regions of an image



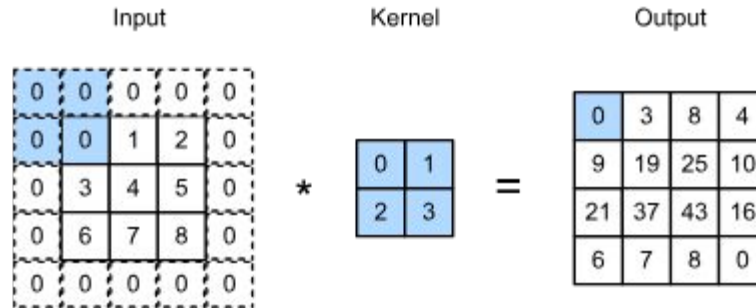
Convolution Layer



Convolution Layer

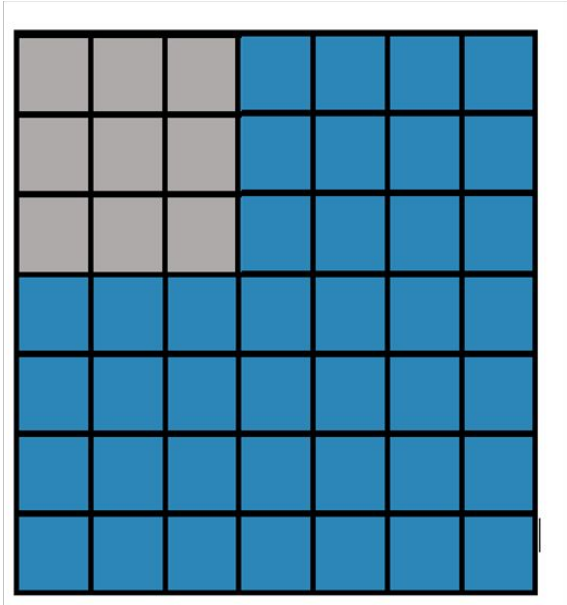


Convolution Layer: Zero Padding

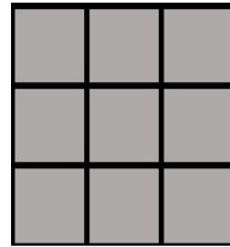


Convolution Layer: Stride

Stride = 2



7x7 input

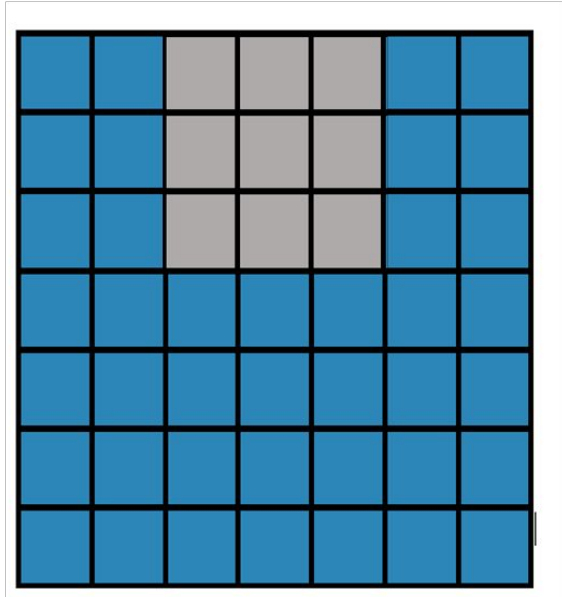


3x3 filter



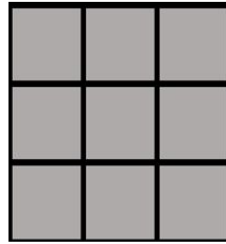
Convolution Layer: Stride

Stride = 2



7x7 input

filters jump 2 pixels at a time as we slide them around

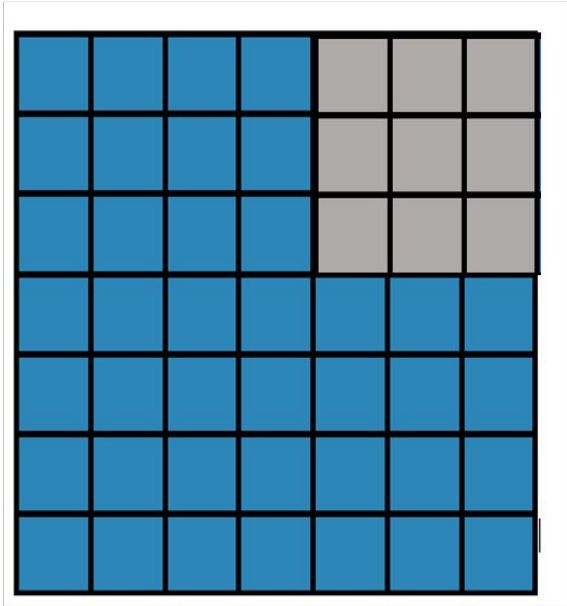


3x3 filter



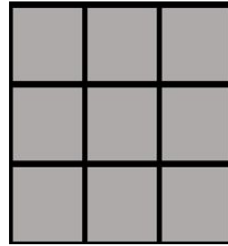
Convolution Layer: Stride

Stride = 2



7x7 input

filters jump 2 pixels at a time as we slide them around

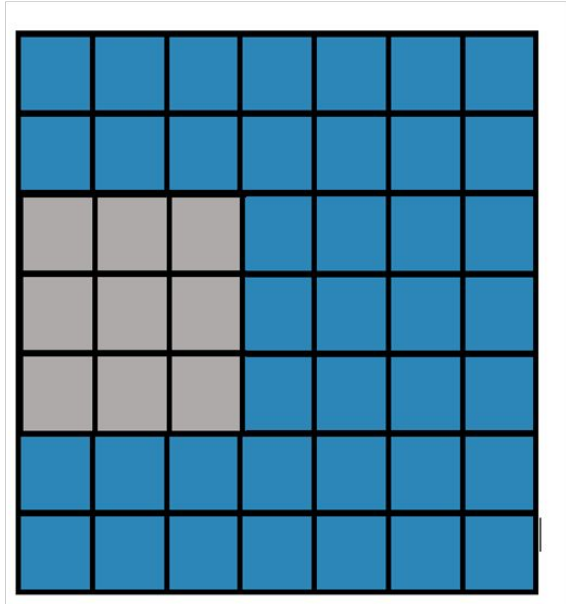


3x3 filter

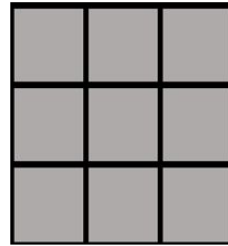


Convolution Layer: Stride

Stride = 2



7x7 input

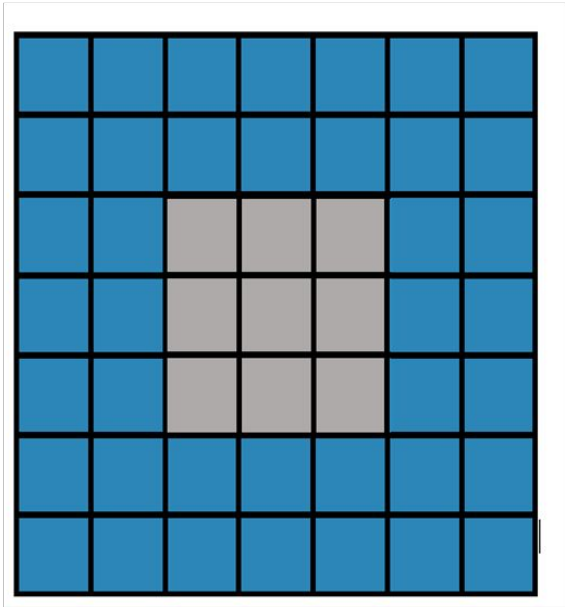


3x3 filter

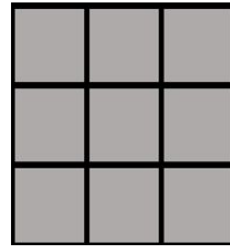


Convolution Layer: Stride

Stride = 2



7x7 input

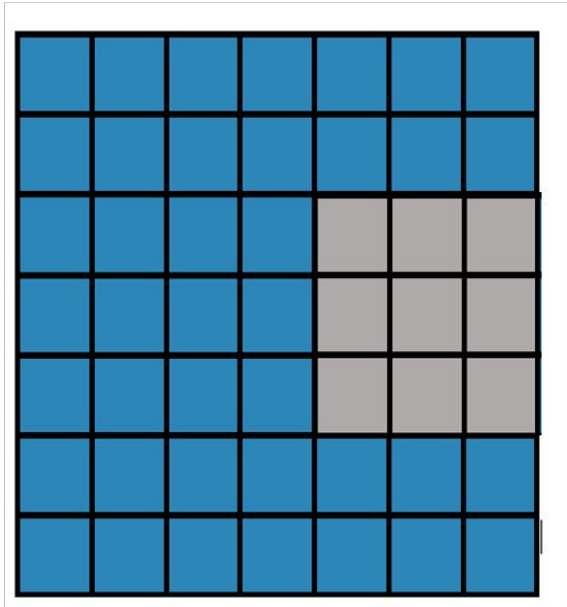


3x3 filter

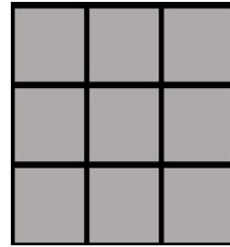


Convolution Layer: Stride

Stride = 2



7x7 input

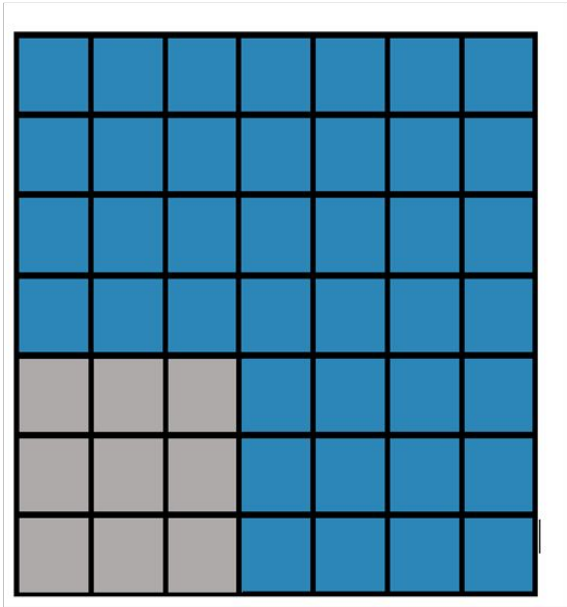


3x3 filter

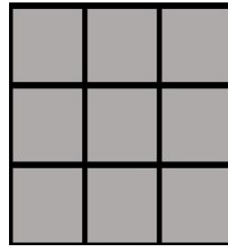


Convolution Layer: Stride

Stride = 2



7x7 input

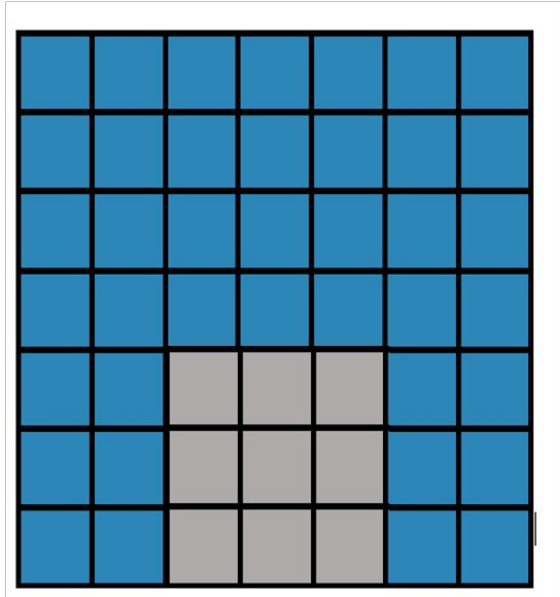


3x3 filter

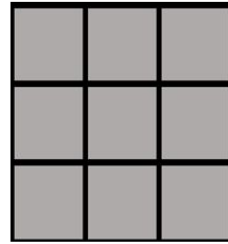


Convolution Layer: Stride

Stride = 2



7x7 input

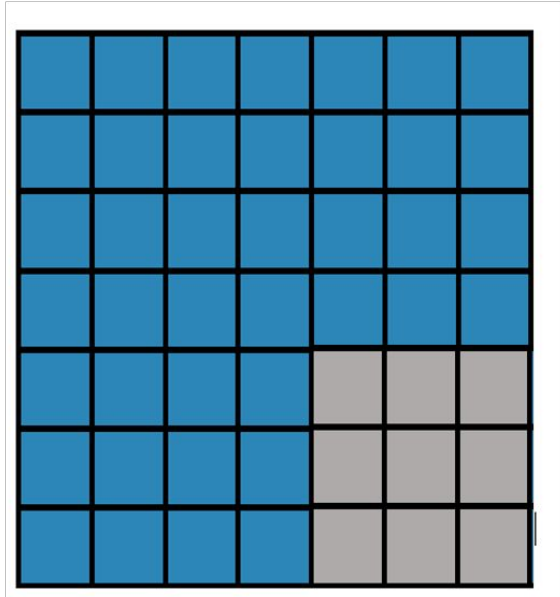


3x3 filter

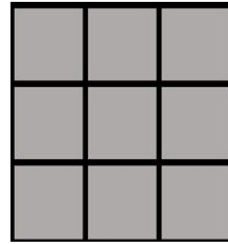


Convolution Layer: Stride

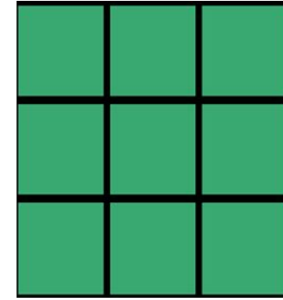
Stride = 2



7x7 input



3x3 filter



3x3 output



Input Size vs Output Size

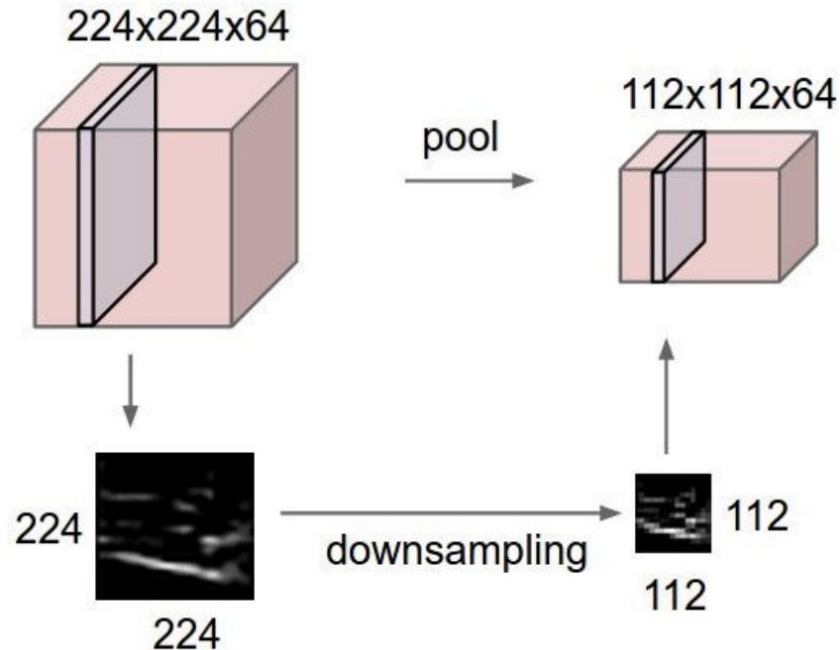
$$\text{output width} = \frac{W - F_w + 2P}{S_w} + 1$$

$$\text{output height} = \frac{H - F_h + 2P}{S_h} + 1$$

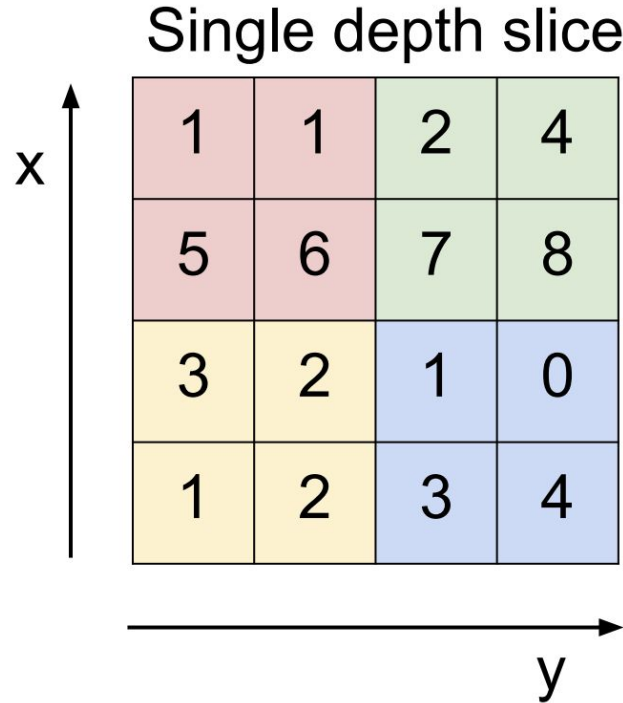


Pooling Layer

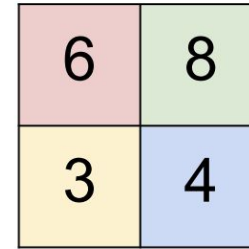
- makes the representations smaller and more manageable
- operates over each activation map independently:



MAX POOLING



max pool with 2x2 filters
and stride 2



Stride is the step size of
moving the filter

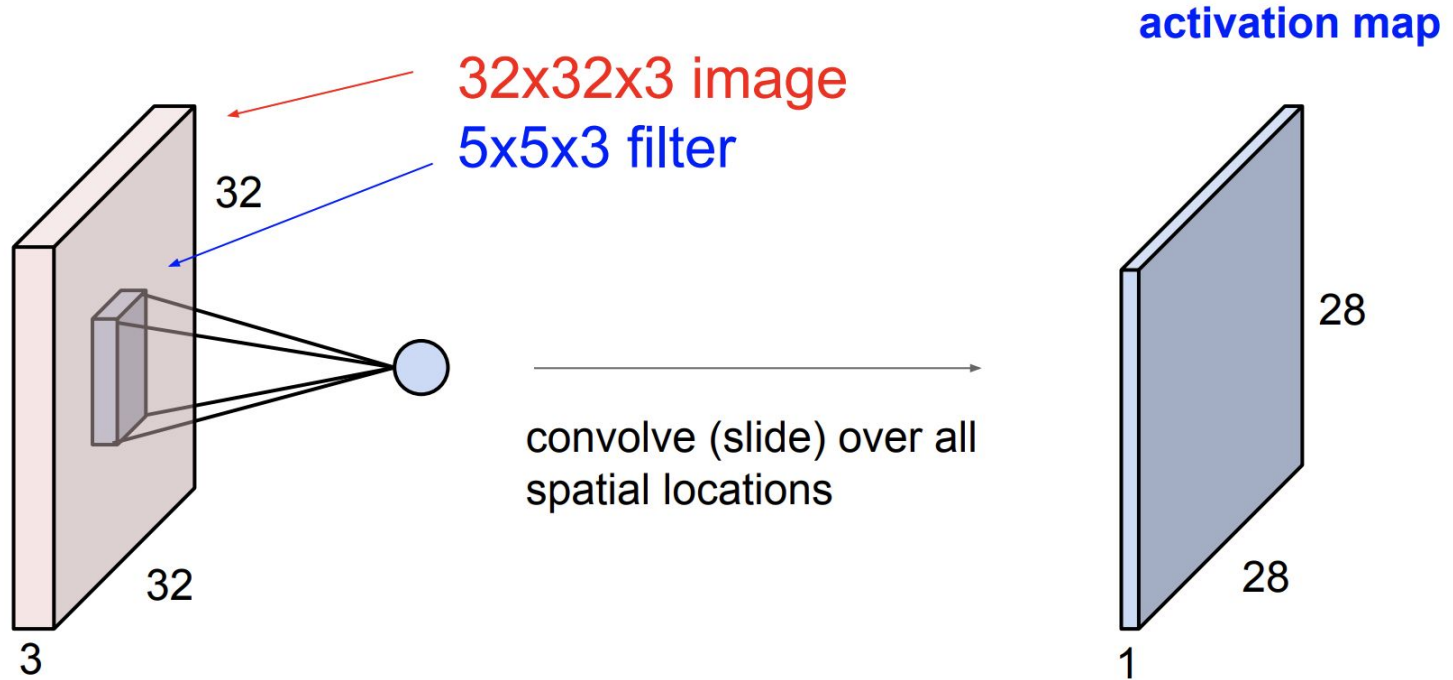


Pooling Layer

- Makes the feature dimension smaller
- Reduce the number of parameters and computation in network; thus control overfitting
- Make the network invariant to small transformations, distortions and translations
- Make the inference almost scale invariant

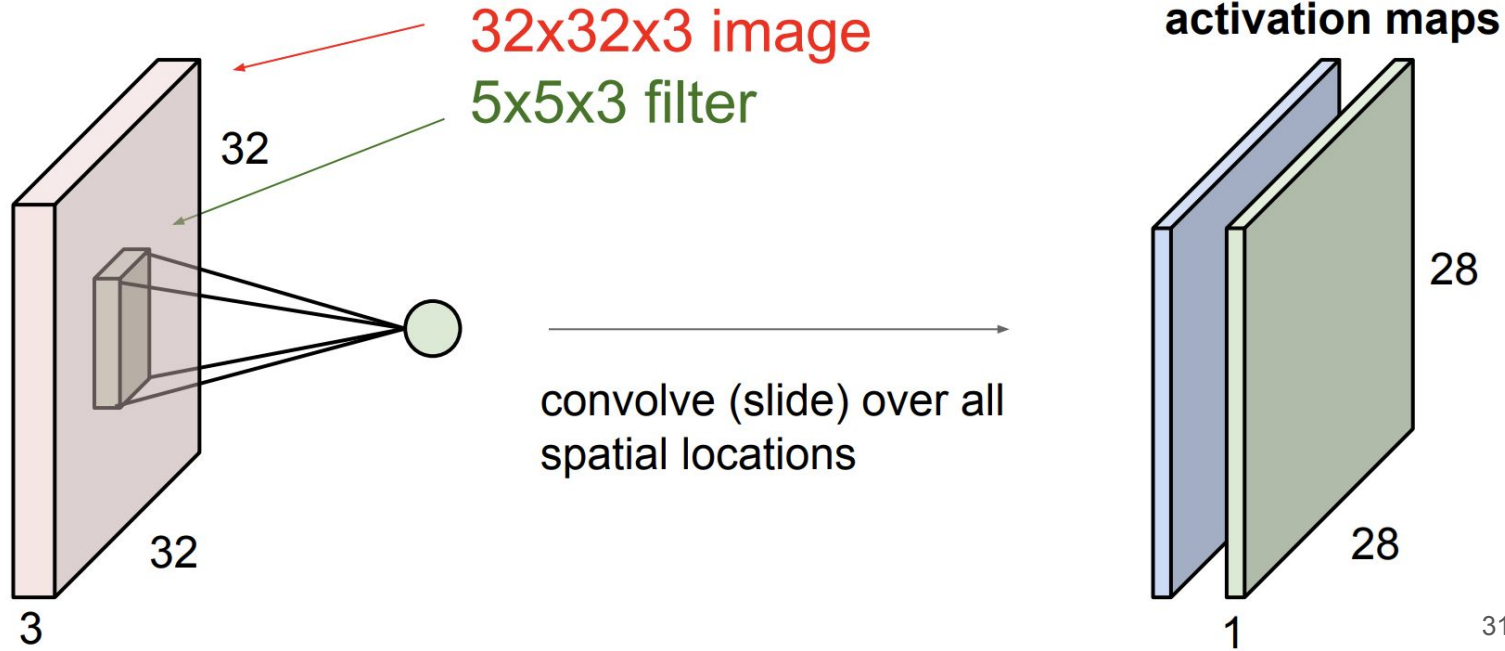


Convolution Layer



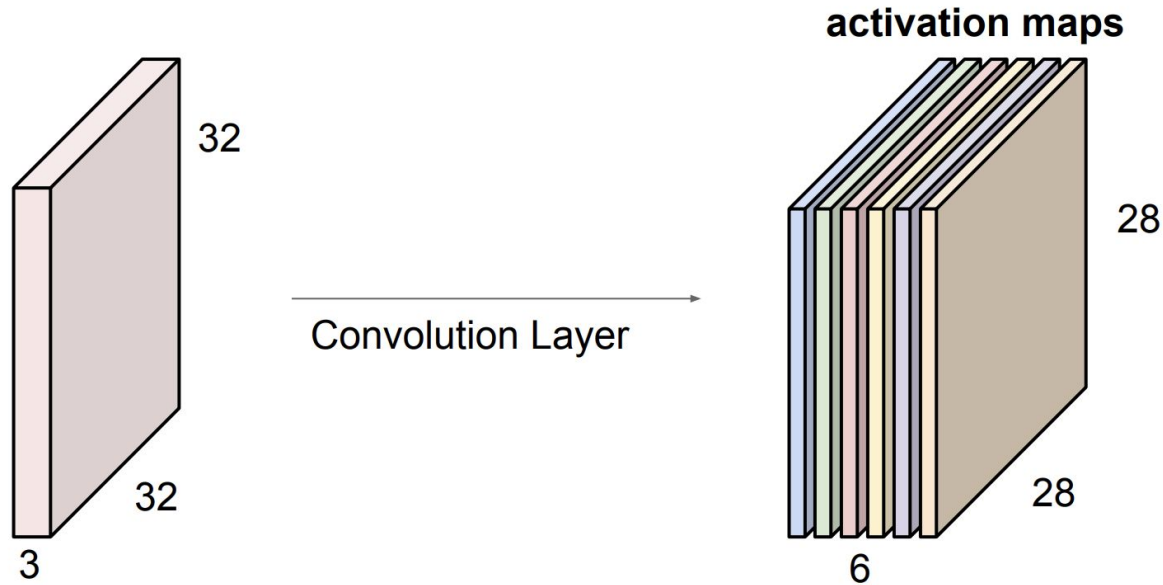
Convolution Layer

consider a second, **green** filter



Convolution Layer

For example, if we had 6 5x5 filters, we'll get 6 separate activation maps:

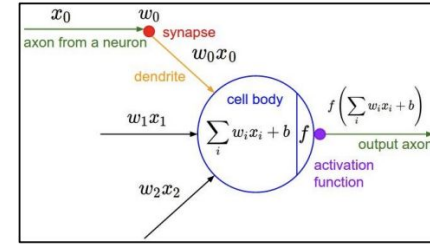
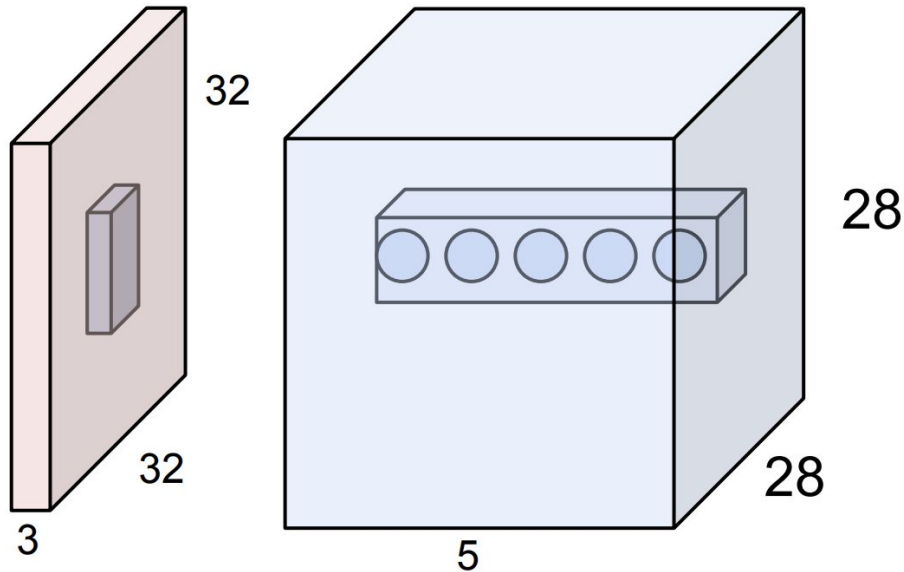


We stack these up to get a “new image” of size 28x28x6!



Convolution Layer

The brain/neuron view of CONV Layer



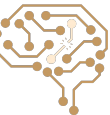
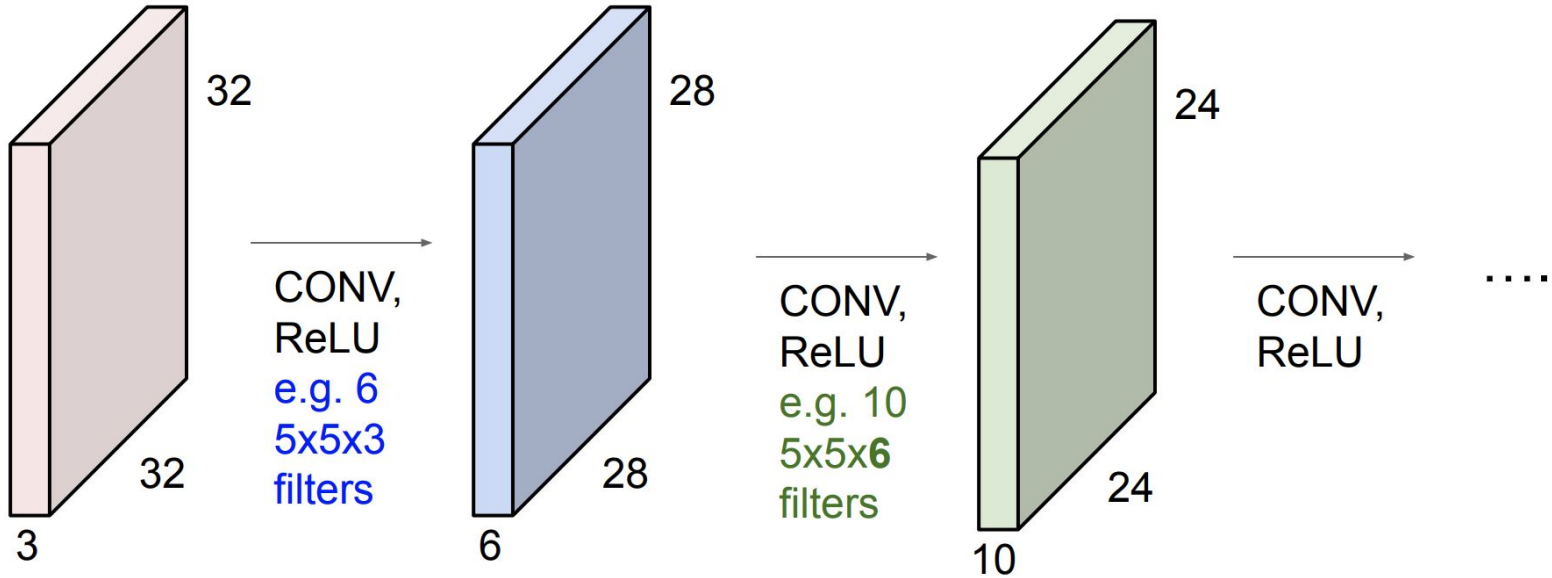
E.g. with 5 filters,
CONV layer consists of
neurons arranged in a 3D grid
(28x28x5)

There will be 5 different
neurons all looking at the same
region in the input volume



Convolution Layer

Preview: ConvNet is a sequence of Convolution Layers, interspersed with activation functions



Sobel Filter

X – Direction Kernel

-1	0	1
-2	0	2
-1	0	1

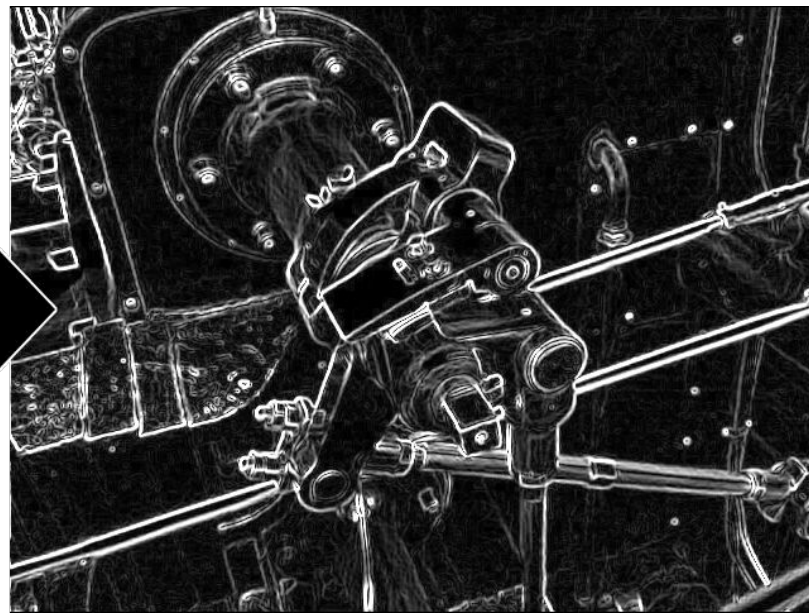
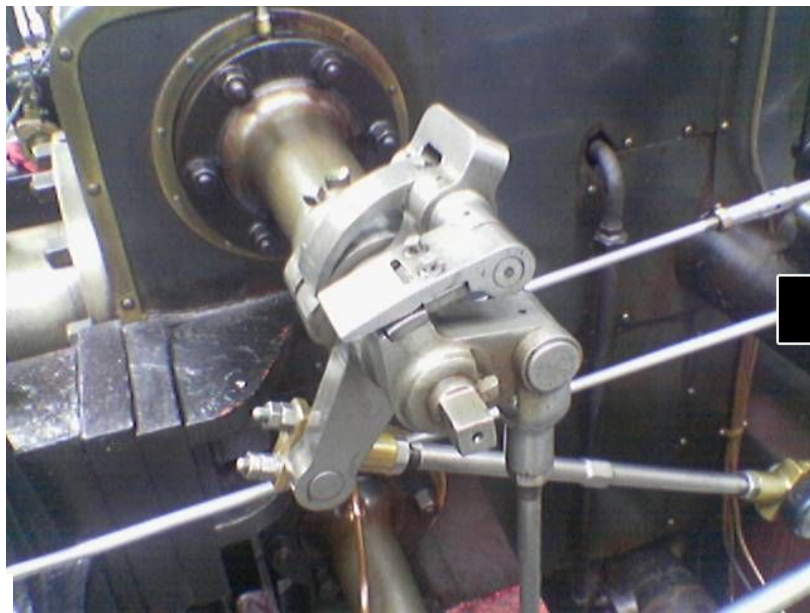
Y – Direction Kernel

-1	-2	-1
0	0	0
1	2	1

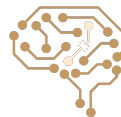
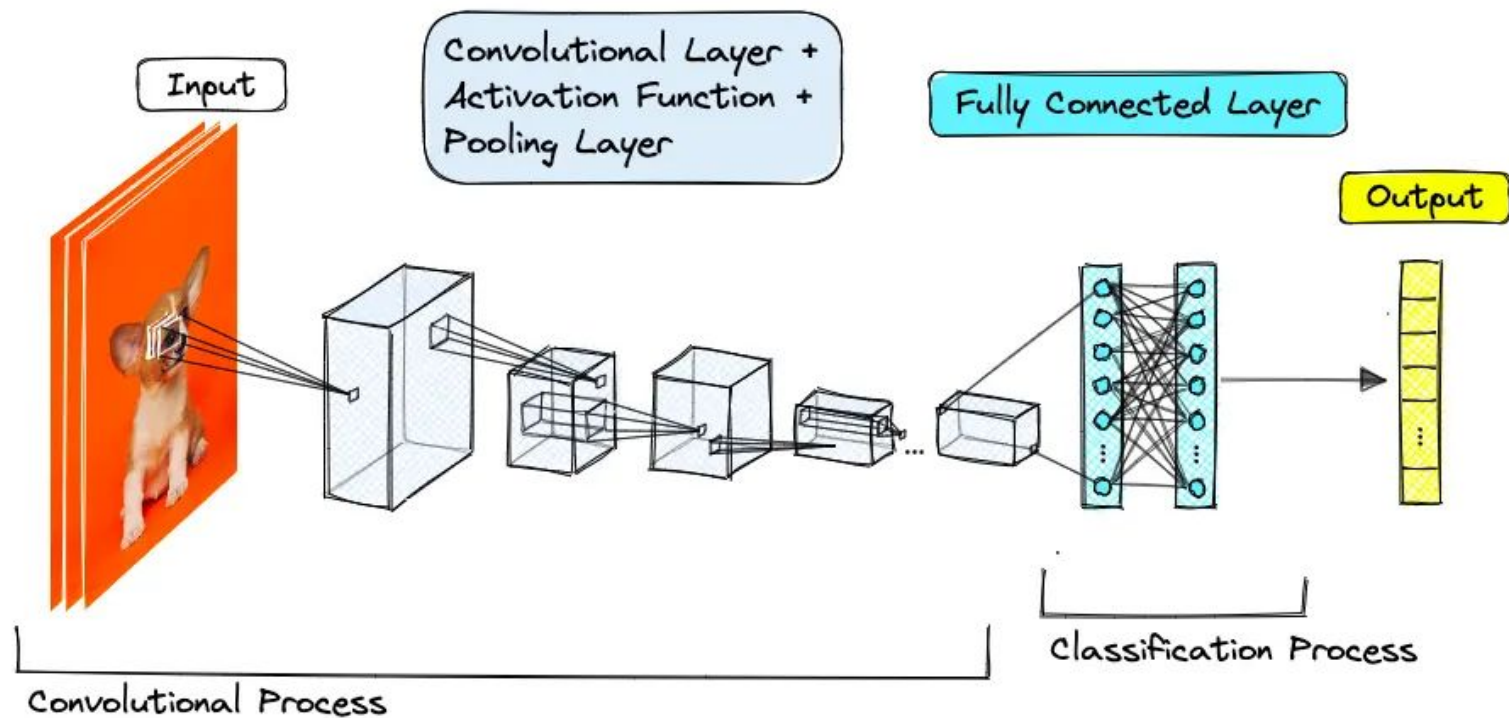


What do convolutional filters learn?

- Edge Detection By Sobel Filter



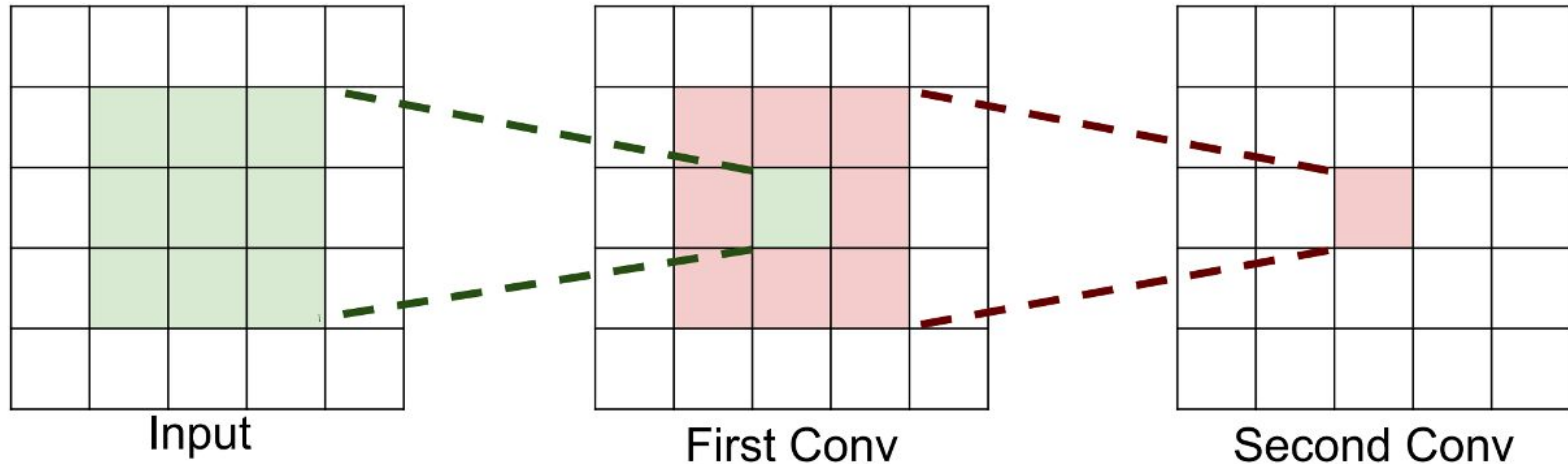
Putting it all together



Receptive Field

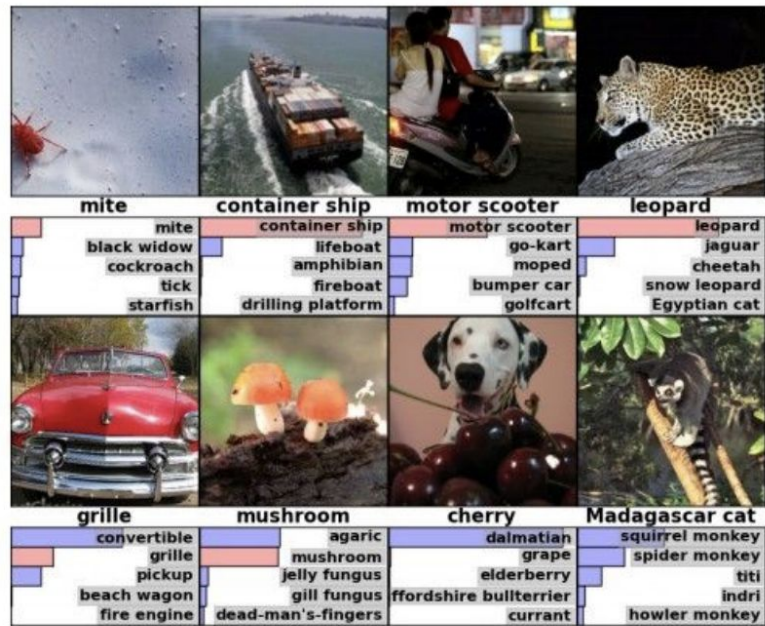
How big of a region in the input does a neuron on the second conv-layer see?

Two 3x3 filters together perform like one 5x5 filter (same receptive field)

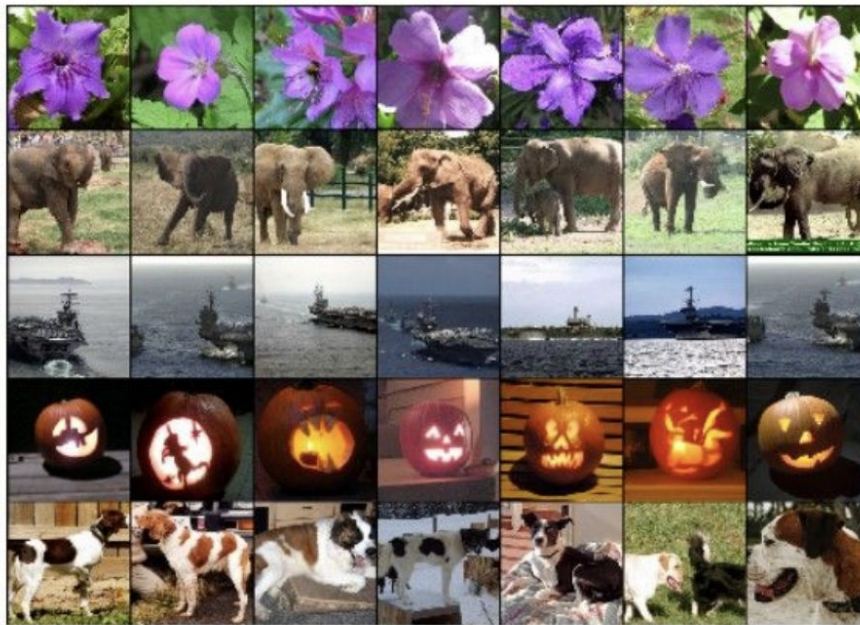


ConvNets are everywhere

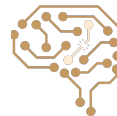
Classification



Retrieval



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ConvNets are everywhere



Photo by Lane McIntosh. Copyright CS231n 2017.

self-driving cars

Photo and figure by Lane McIntosh; not actual example from Mnih and Hinton, 2010 paper.



Mnih and Hinton, 2010



Next Session: RL

